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to local and systemic diseases [3].

Patients in intensive care units (ICUs) present an increased amount of oral biofilm that is susceptible to colonization by multiresistant microorganisms due to the use of multiple antibiotics, drying of the mucous membranes, decreased salivary flow, and accumulation of secretions resulting from the presence of an orotracheal tube [4]. Recent studies indicate that microbial alterations may occur in the oral biofilm of patients under mechanical ventilation [5,6]. These changes predict the colonization of the biofilm by ventilator-associated pneumonia (VAP) associated respiratory pathogens that are not typically found in the oral cavity of healthy individuals [5].

In recent years, research in Adult ICUs has found a high prevalence of respiratory pathogens in saliva and oral biofilm of adult patients under mechanical ventilation [5,7-9]. Recent investigations have been carried out only in adult hospitalized patients under mechanical ventilation [10-16]. Therefore, research in pediatric ICUs is necessary to understand the dynamics of microorganisms in the oral biofilm and how this microbiota relates to the microbiota of tracheal aspirate in children. Thus, this study aimed to investigate the similarity of the microbiota in the tracheal aspirate and oral biofilm of children undergoing mechanical ventilation in a pediatric ICU and to understand the influence of one biofilm on the composition of the other.

■ Material and Methods

Ethical Clearance and Study Design

This research was approved by the Research Ethics Committee of the Presidente Dutra University Hospital – HUUFMA (n° 251.610). It consists of a cross-sectional study carried out at the Pediatric Intensive Care Unit of the University Hospital of the Federal University of Maranhão - Maternal and Child Unit (HUUFMA-UMI) from March to December 2019.

Sample

The sample included 35 children hospitalized (with ages ranging between 1 month and 13 years) on mechanical ventilation via an orotracheal tube. Patients with a previous episode of aspiration of gastric contents, undergoing thoracic surgery, with chronic obstructive pulmonary disease, carriers of the acquired immunodeficiency syndrome virus, with community pneumonia, and those undergoing organ transplants were excluded [9].

Data Collection

A single researcher performed data collection from the medical records of the 35 participants. The data collection included the following variables: gender, age, reason for hospitalization, length of stay, duration of mechanical ventilation, and systemic diseases. A clinical examination was also performed to categorize oral hygiene into satisfactory and unsatisfactory according to the presence of visible lingual biofilm (VLB), adapted from the criteria of the Greene and Vermilion plaque index [17], and for assessment of the oral condition in presence or absence of alterations: teeth (presence caries lesion), gingiva (spontaneous gingival bleeding) and buccal mucosa (erosions, ulcerations, candidiasis).

of tracheal aspirate was performed after collecting the buccal biofilm, following the routine nursing team. After the collections, patients continued to be monitored during their stay in the ICU, with each patient's progress closely followed, until discharge or death.

The lingual biofilm and tracheal aspirate samples were sent to the laboratory, and the culture media used were blood agar, MacConkey agar, Sabouraud agar, and BHI broth. After microbial growth, the inocula were prepared and placed in the Vitek® 2 System (Biomérieux Brasil S.A., Rio de Janeiro, RJ, Brazil) for the identification and performance of the antibiotic sensitivity test (AST) of bacteria and yeasts following the protocols recommended by the manufacturer [18].

Data Analysis

The results were tabulated and analyzed using BioEstat version 5.3 (Optical Digital Technology, Belém, PA, Brazil), employing Fisher's Exact Test to verify the association between the presence of the same microorganism at the two different sites. The significance level adopted was 5% ($p < 0.05$).

■ Results

The study population consisted of 54.28% boys and 45.72% girls, with ages ranging from 1 month to 13 years. The mean age was 51.2 ± 46.8 months, and they were hospitalized in the Pediatric ICU for a minimum of 5 days and a maximum of 69 days, under mechanical ventilation (MV) from 2 to 48 days. The oral condition revealed the presence of gingival bleeding (5.7%), untreated dental caries (20%), soft tissue lesions (17.1%), and unsatisfactory oral hygiene (8.6%). 31.4% of patients died during hospitalization (Table 1).

Table 1. Distribution of demographic and clinical characteristics of the patients admitted to the pediatric intensive care unit.

Variables	Mean	SD	Range	N	%
Sex					
Female				16	45.7
Male				19	54.3
Age Group	51.2	± 46.8	2–156		
Up to 24 months				17	48.6
25 months or more				18	51.4
Reasons for Hospitalization					
Neurologic disorders				14	40.0
Gastrointestinal disorders				6	17.1
Cardiovascular disorders				5	14.3
Nephrological disorders				4	11.5
Other				6	17.1
Length of Hospital Stay (Days)	20.7	± 14.4	5–69		
Up to 20 days				19	54.3
21 days or more				16	45.7
Mechanical Ventilation Length (Days)	12.3	± 10.1	1–48		
Up to 9 days				19	54.3
10 days or more				16	45.7
Gingival Bleeding					
No				33	94.3
Yes				2	5.7
Untreated Dental Caries					

	6	17.1
	32	91.4
Poor	3	8.6
Dentition Group		
Absent	9	25.7
Primary	13	37.1
Mixed	10	28.6
Permanent	3	8.6
Hospital Outcome		
Discharge	24	68.6
Death	11	31.4

SD: Standard Deviation.

Eighteen different species were identified in the lingual biofilm and 14 different species in the tracheal aspirate (Figure 1). The most frequent species in the lingual biofilm were *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, *Klebsiella pneumoniae*, *Staphylococcus aureus*, and *Stenotrophomonas maltophilia*. In tracheal aspirate, the most frequent species were *Acinetobacter baumannii* and *Stenotrophomonas maltophilia*. The same species of microorganism was identified in 12 patients (34.3%). The highest frequencies of the same species isolated in both regions were *Acinetobacter baumannii* (3 patients, 8.6%), *Stenotrophomonas maltophilia* (3 patients, 8.6%), and *Pseudomonas aeruginosa* (2 patients, 5.7%).

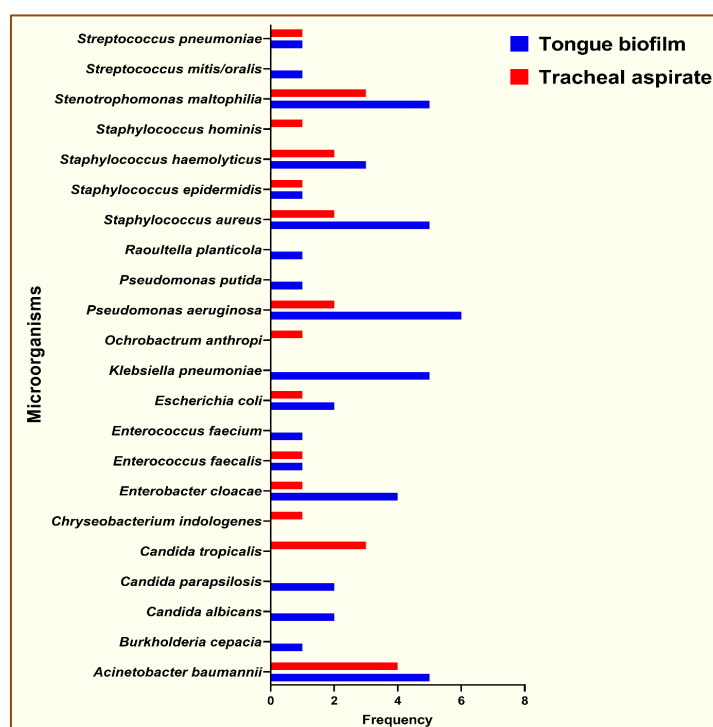


Figure 1. Absolute frequency of microorganism species isolated from lingual biofilm and tracheal aspirate.

The frequency of the same species at both sites in patients aged up to 24 months was statistically higher compared to the older group (58.8% versus 11.1%, $p = 0.004$); other variables did not show statistically significant

phic and clinical factors associated with the detection of the same microorganism in and tracheal samples.

Variables	Same Microorganism in the Tongue Biofilm and Tracheal Samples				p-value
	N	Yes %	N	No %	
Sex					0.478
Female	4	25.0	12	75.0	
Male	8	42.1	11	57.9	
Age Group					0.004*
Up to 24 months	10	58.8	7	41.2	
25 months or more	2	11.1	16	88.9	
Length of Hospital Stay (Days)					0.728
Up to 20 days	7	36.8	12	63.2	
21 days or more	5	31.2	11	68.8	
Mechanical Ventilation Length (Days)					0.279
Up to 9 days	5	26.3	14	73.7	
10 days or more	7	43.8	9	56.2	
Gingival Bleeding					1.000
No	11	34.4	21	65.6	
Yes	1	50.0	1	50.0	
Untreated Dental Caries					0.069
No	12	42.9	16	57.1	
Yes	0	0	7	100	
Oral Mucosal Lesion					1.000
No	10	34.5	19	65.5	
Yes	2	33.3	4	66.7	
Oral Hygiene					1.000
Good/regular	11	34.4	21	65.6	
Poor	1	33.3	2	66.7	
Dentition Group					0.181
Absent	5	55.6	4	44.4	
Primary	5	38.5	8	61.5	
Mixed	1	10.0	9	90.0	
Permanent	1	33.3	2	66.7	
Hospital Outcome					0.087
Discharge	6	25.0	18	75.9	
Death	6	54.6	5	45.4	

*Statistically Significant.

Demographic and clinical factors associated with hospital death were investigated (Table 3). There was a statistically higher frequency of deaths among patients aged up to 24 months (52.9% *versus* 11.1%, $p = 0.01$). In addition, there was a higher frequency of death among patients with MV time ≥ 10 days, but without significant differences (50% *versus* 15.8%, $p = 0.065$).

Table 3. Distribution of demographic and clinical factors according to hospital outcome.

Variables	Hospital Outcome				p-value
	Discharge N	Discharge %	Death N	Death %	
Sex					0.492
Female	12	75.0	4	25.0	
Male	12	63.2	7	36.8	
Age Group					0.011*

	10	62.5	6	37.5	
ion Length (Days)					0.065
	16	84.2	3	15.8	
10 days or more	8	50.0	8	50.0	
Gingival Bleeding					0.536
No	23	69.7	10	30.3	
Yes	1	50.0	1	50.0	
Untreated Dental Caries					0.391
No	18	64.3	10	35.7	
Yes	6	85.7	1	14.3	
Oral Mucosal Lesion					0.639
No	19	65.5	10	34.5	
Yes	5	83.3	1	16.7	
Oral Hygiene					1.000
Good/regular	22	68.8	10	31.2	
Poor	2	66.7	1	33.3	
Dentition Group					0.137
Absent	5	55.6	4	44.4	
Primary	7	53.8	6	46.2	
Mixed	9	90.0	1	10.0	
Permanent	3	100	0	0	

*Statistically Significant.

■ Discussion

This study revealed that the same bacteria were present in the lingual biofilm and in the tracheal aspirate, and were identified in 34.3% of patients undergoing invasive mechanical ventilation in a pediatric intensive care unit. A similar result was demonstrated by Souza et al. [9], who reported the same species of pathogens in tracheal aspirate and buccal biofilm in 59.37% of adult patients on mechanical ventilation. However, there was a lower frequency of the same bacterial species in the lingual biofilm and in the tracheal aspirate of patients hospitalized in the Pediatric ICU compared to those in the Adult ICU.

In the present study, 18 different species of microorganisms were identified in the lingual biofilm, and 14 other species were identified in the tracheal aspirate. In another study, only eight different species were detected in the lingual biofilm, and four in the tracheal aspirate. This difference can be attributed to the sample size, as the present study evaluated 35 children on mechanical ventilation, whereas the other study included only eight children [6].

Acinetobacter baumannii and *Stenotrophomonas maltophilia* were the most frequent bacterial species, simultaneously, in both sites. *Acinetobacter baumannii* is a Gram-negative coccobacillus associated with persistent nosocomial infections [19], which cause morbidity and mortality, especially in intensive care units [20]. Although *A. baumannii* has been frequently reported as the primary pathogen in adults, it is also a critical concern in the pediatric intensive care unit, particularly in cases of prolonged hospitalization among pediatric patients [21].

In our study, three patients had *A. baumannii* in the two investigated sites, and 2 of them died while 1 developed a bloodstream infection. Another survey found that mortality (16.7%) and positive blood cultures for *A. baumannii* infection (27.5%) were associated with this pathogen in a Pediatric ICU [22].

Stenotrophomonas maltophilia is a gram-negative, aerobic, and non-fermentative bacterium [19]. Due to its low virulence, it rarely causes disease in healthy individuals; however, in immunosuppressed individuals, it

n in the aspirate, in 5 and 3 participants, respectively. These data were not published in the respiratory tract was defined as the primary site of infection by *S. maltophilia*.

Mortality related to *S. maltophilia* is significantly high, both in adults (37.5%) and in children, with rates ranging from 10% to 60% [24,26-28]. In the present study, two children who simultaneously exhibited *S. maltophilia* in the lingual biofilm and tracheal aspirate died, and one of these children developed ventilator-associated pneumonia.

The rate of VAP in the sample was minimal, with only one patient diagnosed in this study. This can be attributed to the presence of a dentist in the Pediatric ICU, integrated into the patient care routine through a service that has been in place for 5 years, and to the strict hygiene protocol instituted, which utilizes a 0.12% chlorhexidine solution twice a day to reduce the microbial load in the oral cavity. The use of oral hygiene protocols, including mouthwashes or chlorhexidine gel, was associated with a 40% reduction in the incidence of ventilator-associated pneumonia [29]. However, some studies mention that the use of chlorhexidine does not decrease the incidence of VAP in children on mechanical ventilation [30,31].

Only one patient in the Pediatric ICU (2.85%), among the 35 participants, did not present pathogens in the lingual biofilm, and another respiratory bacterium frequently related to VAP, *Pseudomonas aeruginosa*, was identified in oral biofilm samples, corroborating with other studies [5,8,9], which leads us to conclude that the mouth is a reservoir of respiratory pathogens.

There was a higher frequency of deaths among patients aged up to 24 months (52.9% versus 11.1%, $p = 0.01$). The newborn's immune system presents a limited response against invasive pathogens and greater susceptibility to infections [32]. Additionally, the production of secretory IgA is absent in the first days of life, causing increased vulnerability to pathogenic microorganism invasions [33]. Furthermore, the antibody response to polysaccharides present in the cell membrane of bacteria occurs in the marginal zone of the spleen. Histological studies of the infant spleen have shown that this zone is not fully developed until 2 years of age, which may, in part, explain the delayed response of antibodies to encapsulated bacteria, such as *Acinetobacter baumannii* and *Pseudomonas aeruginosa* [34].

The number of participants in this study can be considered small; however, it is justified by the low patient turnover and the exclusion of children with severe conditions from the research. However, in the present research, biological material was collected from two different sites of the same patient in a pediatric ICU for comparison - a method not employed in other studies, where only one site was investigated. In addition, the present study employed the same technique to identify microorganisms from two sites: lingual biofilm and aspirate, thus minimizing possible reading errors and making it a pioneering study in the investigation of pathogens in lingual biofilm of patients in the Pediatric ICU.

■ Conclusion

There was a high similarity in the microorganisms present in the lingual biofilm and tracheal aspirate of patients up to 24 months of age, and this similarity was greater among patients who died. Thus, it is noted that the oral cavity of children hospitalized on invasive mechanical ventilation serves as a reservoir of respiratory pathogens, reiterating the need for Dental Care in all Intensive Care Units.

■ Financial Support

None.

■ Conflict of Interest

The authors declare that they have no conflicts of interest.

■ Data Availability

■ References

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